large electron density and dissolved band structure electron-ion recombination dominates [156]. At irradiance values leading to optical breakdown with cavitation bubble formation, cross recombination processes between electrons and holes or electrons and ions from independent ionization events dominates by far over geminate recombination [157]. Under these circumstances, recombination is proportional to the square of free-carrier density because two types of free-carriers are involved in each event [113,158]. The recombination time ranges from ≈ 1ps to several ps, depending on ρ_c [37]. Therefore, recombination influences the breakdown dynamics in ps and ns breakdown but is negligible for fs pulses.

Thermalization of the absorbed laser energy involves energy transfer from free electrons to heavy particles occurring through elastic and inelastic collisions [159-161] as well as through recombination followed by energy dissipation via vibrational relaxation [162-164]. Thermalization through inelastic collisions with vibrational excitation of water molecules and dissipation through the hydrogen bond network is fast and dominates for low-density plasmas such as the electron cascade produced by ionizing radiation and occurs on a time scale below 1 ps [165-168]. Experimental investigations yielded plasma life times ≤ 300 fs for fs plasmas with low electron densities ≤ 3 x 10^{18} cm^{-3} [169,170], two orders below the bubble threshold [37]. In low-density plasma, hydrating CB electrons interact with intact water molecules and are then solvated in traps below the CB [171]. By contrast, in high-density plasmas well above the breakdown threshold, water molecules are largely dissociated and the band structure is disturbed [156,172,173]. In this regime, electron collisions are mainly elastic, and longer thermalization times, τ_{therm}, in the order of 20-30 ps have been observed [123,174]. Here, thermalization proceeds mainly via electron-ion recombination. The recombination rate for high-density plasmas was determined by analyzing the decay of plasma scattering [123,174] and luminescence [158]. Before thermalization, the electron temperature exceeds the ion or lattice temperature, respectively, and during the thermalization process both temperatures equilibrate and a thermodynamic temperature evolves [152,156,175-178]. The thermodynamic temperature then governs the onset of hydrodynamic phenomena such as shock wave emission and bubble formation [7,153,179].

Heating of the focal volume during the laser pulse causes a gradual change of the electronic energy distribution from Fermi-Dirac statistics towards a Boltzmann distribution [159,180,181]. The change of the energy distribution can significantly contribute to the CB electron density, when high temperatures are reached and the high-energy tail of the Boltzmann distribution reaches into the CB. We denote this temperature-related process as "*thermal ionization*." Since it refers to the electron density distribution in

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thermal equilibrium between electron and heavy particle populations, it differs significantly from strong-field and avalanche ionization, which create a thermal imbalance by depositing energy into the electronic system. As long as the band structure is still intact, the temperature-related density of electrons in the conduction band, ρ_{therm}, can be assessed using the theory of electric conductivity of semiconductors [180,181]. When the band structure is dissolved at high temperatures, the ionization degree is described by the Saha equation [182-184]. Both approaches predict a proportionality $\rho_{therm} \propto exp(-E_{gap}/2k_B T)$, where E_{gap} is the band gap energy, and k_B the Boltzmann constant [159,180-183]. The exponential increase of ρ_{therm} with T suggests a rapid onset of thermal ionization beyond a critical temperature T_{cr}, which will become relevant at times $t > \tau_{therm}$.

With intact band structure, ionization will cease, when ρ_c reaches such a large value that the highest molecular orbital of the valence band is depleted [6,143,156,185]. This level is the $1b_1$ level, which contains a pair of bound electrons [185,186]. Since the molecular density of liquid water is 3.34×10^{22} cm^{-3}, full depletion of the $1b_1$ level corresponds to a free electron density of $\rho_c = 6.68\times10^{22}$ cm^{-3} [113]. However, electron densities that fully deplete the $1b_1$ level are likely associated with a dissolution of the band structure of water and dissociation of water molecules leading to the development of an electron-ion plasma. In any case, once a very high ionization degree and electron density are reached, most absorbed laser energy goes into inverse bremsstrahlung absorption with little further impact ionization, which results in rapid heating of the free electrons to high levels of kinetic energy [125].

Electron and heat diffusion out of the focal volume inhibit the breakdown process both during surface ablation of transparent dielectrics and plasma formation in bulk media but on very different time scales. During surface ablation, a thin plasma skin layer with nanometer thickness evolves, and electron and heat diffusion become relevant already on a time scale of a few picoseconds [125,137,139,177,178,187]. The heat diffusion time is proportional to the square of the thickness of the absorbing layer, and the heat penetration depth is proportional to $(D\cdot t)^{1/2}$, where D is the heat diffusion coefficient and t the time [187,188]. Hence, the heated zone spreads by a factor of $\tau_L^{1/2}$ during the laser pulse, which must be compensated by an increase of radiant exposure (fluence) by the same factor to achieve the ablation temperature. This leads to a proportionality of breakdown fluence to $\tau_L^{1/2}$ both for dielectrics and metals [112,125,177,187]. By contrast, for breakdown in bulk dielectrics, no skin layer evolves [133,134]. Even with tight focusing, focal dimensions are in the range of hundreds of nanometers to several micrometers, and diffusion becomes relevant only on a time scale of hundreds of nanoseconds [22,156,188,189].

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Therefore, electron diffusion and heat conduction can be neglected. Here the breakdown threshold for pulse durations > 10 ps depends mainly on the irradiance at which multiphoton-generated seed electrons for AI are available [22].

The above considerations reveal that essentially four interaction mechanisms acting on different time scales govern the evolution of CB electron density in bulk water. These are strong-field ionization SI, avalanche ionization AI, recombination, and thermal ionization TI. While fs breakdown can be understood by analyzing only the interplay of SFI and AI, for longer pulse durations also recombination and thermal ionization must be considered. This brings in two new feedback loops (Fig. 1), which complicates the interplay of the basic mechanisms. For getting a general understanding of the breakdown behavior in the entire pulse duration range from fs to ns, it is useful to keep the description of the individual mechanisms as simple as possible. Therefore, we describe their dependencies on irradiance I, CB electron density ρ_c and temperature T by a set of generic equations:

$$\left(\frac{d\rho_c}{dt}\right)_{MPI} = \eta_{MPI}\, I^k\,, \tag{1}$$

$$\left(\frac{d\rho_c}{dt}\right)_{AI} = \eta_{AI}\, \rho_c\, I\,, \tag{2}$$

$$\left(\frac{d\rho_c}{dt}\right)_{rec} = -\eta_{rec} \times \rho_c^2\,, \tag{3}$$

$$\rho_{c,therm} \propto e^{-E_{gap}/2k_BT}\,. \tag{4}$$

For the sake of simplicity, we consider here only multiphoton ionization as representative for strong-field ionization and use the simple AI rate equation for the stationary avalanche ionization regime. For the sake of generality, we assume a simple band gap without energy levels between valence and conduction band. The actual band structure is more complex in water, where an intermediate state at the energy level of solvated electrons facilitates plasma initiation [36,37,149], and in fused silica, where self-trapping of excitons impairs avalanche ionization [190,191]. Furthermore, we assume a constant band gap during plasma formation although the band structure deteriorates once high electron densities and temperatures are reached in the focal volume, and the rate constants for all ionization processes change.

The relative strength of MPI and AI depends on laser wavelength and pulse duration, as discussed further above for fs breakdown. The strength of recombination and thermal ionization compared to MPI and AI depends mainly on pulse duration and will be discussed in detail in the next section.

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